

Cryptology Assignment-3

1. Determine the multiplicative inverse modulo 26 for each of the following integers:

1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25 mod 26.

Ans:

$$1 \rightarrow 1 \times 1 = 1 \equiv 1 \pmod{26}$$

$$26 \times 1 = 26$$

$$26 \times 2 = 52$$

$$26 \times 3 = 78$$

$$3^{-1} \rightarrow 26 + 1 = 27 \div 3 \equiv 9$$

$$26 \times 4 = 104$$

$$5^{-1} \rightarrow 104 + 1 = 105 \div 5 \equiv 21$$

$$26 \times 5 = 130$$

$$7^{-1} \rightarrow 104 + 1 = 105 \div 7 \equiv 15$$

$$26 \times 6 = 156$$

$$9^{-1} \rightarrow 27 \div 3 \equiv 9$$

$$26 \times 7 = 182$$

$$11^{-1} \rightarrow 208 + 1 = 209 \div 11 \equiv 19$$

$$26 \times 8 = 208$$

$$15^{-1} \rightarrow 104 + 1 = 105 \div 15 \equiv 7$$

$$26 \times 9 = 234$$

$$17^{-1} \rightarrow 390 + 1 = 391 \div 17 \equiv 23$$

$$26 \times 10 = 260$$

$$19^{-1} \rightarrow 208 + 1 = 209 \div 19 \equiv 11$$

$$21^{-1} \rightarrow 104 + 1 = 105 \div 21 \equiv 5$$

$$23^{-1} \rightarrow 390 + 1 = 391 \div 23 \equiv 17$$

$$25^{-1} \rightarrow 624 + 1 = 625 \div 25 \equiv 25$$

2. Choose any string of your choice to encrypt using the affine cipher with key values: multiplier 5 and additive 8.

Given $K_1 = 5$, $K_2 = 8$

Ans: For Encryption,

$$C = E(K_1 * P + K_2) \pmod{26}$$

Plaintext, $K_1 \times P$, $+K_2$, mod 26, Output.

H	7	$5 \times 7 = 35 + 8$	43	17	R
E	4	$5 \times 4 = 20 + 8$	28	2	C
L	11	$5 \times 11 = 55 + 8$	63	11	L
L	11	$5 \times 11 = 55 + 8$	63	11	L
O	14	$5 \times 14 = 70 + 8$	78	0	A

3. Encrypt "HI" using the Affine Cipher with $a = 11$ and $b = 5$.

Ans: Given $K_1 = 11$, $K_2 = 5$
 $C = E(K_1 * P + K_2) \bmod 26$

Plaintext		$K_1 * P + K_2$	$\bmod 26$	Output
H	7	$7 * 11 = 77 + 5 = 82$	4	E
I	8	$8 * 11 = 88 + 5 = 93$	15	P

Cipher Text: EP

4. The cipher UCR was encrypted using the affine function $9x + 2 \bmod 26$. Find the plaintext.

Ans: For decryption,

$$P = K_1^{-1} (C - K_2) \bmod 26$$

$$K_1 = 9 \quad K_2 = 2$$

$$\Rightarrow 9^{-1} \bmod 26 = 3$$

Ciphertext		$K_1^{-1} (C - K_2) \bmod 26$	Plaintext
U	20	$3(20 - 2) = 54$	C
C	2	$3(2 - 2) = 0$	A
R	17	$3(17 - 2) = 45$	T

Plain Text: CAT

5. Decrypt the cipher text using Affine Cipher, KDYXJUNWDEXZY using the key 3 (multiplier) and 4 (Additive).

P. T. O.

Ans: For Decry $P = K_1^{-1}(C - K_2) \bmod 26$

Given, $K_1 = 3$, $K_2 = 4$

$$\Rightarrow 3^{-1} \bmod 26 = 9$$

Cipher Text		$K_1^{-1}(C - K_2) \bmod 26$	Plain text
K	10	$9(10 - 4) = 54 \bmod 26 = 2$	C
D	3	$9(3 - 4) = -9 \bmod 26 = 17$	R
Y	24	$9(24 - 4) = 180 \bmod 26 = 24$	Y
X	23	$9(23 - 4) = 171 \bmod 26 = 15$	P
J	9	$9(9 - 4) = 45 \bmod 26 = 19$	T
U	20	$9(20 - 4) = 144 \bmod 26 = 14$	O
W	22	$9(22 - 4) = 162 \bmod 26 = 6$	G
D	3	$9(3 - 4) = -9 \bmod 26 = 17$	R
E	4	$9(4 - 4) = 0 \bmod 26 = 0$	A
X	23	$9(23 - 4) = 171 \bmod 26 = 15$	P
Z	25	$9(25 - 4) = 189 \bmod 26 = 7$	H
Y	24	$9(24 - 4) = 180 \bmod 26 = 24$	Y

Plain Text: CRYPTOGRAPHY.

6. Use $K_1 = 13$, $K_2 = 4$ (Even though it is not valid key-try to encrypt). Encrypt the string "input" and "after" using the above key and find the issue.

Ans: For Encryption,

$$C = (K_1 * P + K_2) \bmod 26$$

Given, $K_1 = 13$, $K_2 = 4$.

Plaintext		$K_1 * P + K_2 \bmod 26$	output
I	8	$13 \times 8 + 4 = 108 \bmod 26 = 4$	E
N	13	$13 \times 13 + 4 = 173 \bmod 26 = 17$	R
P	15	$13 \times 15 + 4 = 199 \bmod 26 = 17$	R
U	20	$13 \times 20 + 4 = 264 \bmod 26 = 4$	E
T	19	$13 \times 19 + 4 = 251 \bmod 26 = 17$	R

A	0	$13 \times 0 = 0 + 4 = 4$	4	E
L	11	$13 \times 11 = 143 + 4 = 147$	17	R
T	19	$13 \times 19 = 247 + 4 = 251$	17	R
E	4	$13 \times 4 = 52 + 4 = 56$	4	E
R	17	$13 \times 17 = 221 + 4 = 225$	17	R

Issue: $K_1 = 13$ (Multiplier) is not a coprime with 26.

The encryption is not one-to-one, and proper decryption is impossible.

7. Find the Inverse matrix for the key $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \pmod{26}$.

Ans: $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \Rightarrow (3 \times 7) - (2 \times 5) = 21 - 10 = \underline{\underline{11}}$

$\Rightarrow 11^{-1} \pmod{26} = 19$

$19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \Rightarrow \begin{bmatrix} 133 & -38 \\ -95 & 57 \end{bmatrix} \pmod{26}$

$= \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$

$K^{-1} = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$

8. Find the inverse of the following matrix whose entries are considered modulo 26:

$\begin{bmatrix} 11 & 13 \\ 2 & 3 \end{bmatrix}$

Ans: $\begin{bmatrix} 1 & 1 & 13 \\ 2 & & 3 \end{bmatrix} \Rightarrow (11 \times 3) - (13 \times 2) = 33 - 26 = 7$

$$\Rightarrow 7^{-1} \bmod 26 = 15.$$

$$15 \begin{bmatrix} 1 & 13 \\ 2 & 3 \end{bmatrix} \Rightarrow \begin{bmatrix} 15 & 195 \\ -30 & 45 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 19 & 13 \\ 22 & 9 \end{bmatrix}$$

9. Is the key valid?

$$\begin{bmatrix} 2 & 14 \\ 13 & 21 \end{bmatrix}$$

Ans:

$$\begin{bmatrix} 2 & 14 \\ 13 & 21 \end{bmatrix} \Rightarrow (2 \times 21) - (14 \times 13) = 42 - 182 = -140$$

$\Rightarrow$ "140" is not a coprime with 26.

Key Invalid!

10. Encrypt using 2x2 Hill Cipher for the plaintext "SOLVED" using $\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix}$

Ans:

$$\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} \Rightarrow (3 \times 11) - (8 \times 7) = (33 - 56) = -23$$

$\Rightarrow 23^*$ is co-prime with 26

Plain Text: "S O L V E D"

18 14 11 21 4 3

i) $P = \begin{bmatrix} 18 \\ 14 \end{bmatrix} = \begin{bmatrix} S \\ O \end{bmatrix} \Rightarrow K = \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix}$

$$\Rightarrow \begin{bmatrix} 18 \\ 14 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} = \begin{bmatrix} 166 \\ 280 \end{bmatrix} \bmod 26 = \begin{bmatrix} 10 \\ 20 \end{bmatrix} = \begin{bmatrix} K \\ U \end{bmatrix}$$

$$(ii) \quad P = \begin{bmatrix} L \\ V \end{bmatrix} = \begin{bmatrix} 11 \\ 21 \end{bmatrix} \Rightarrow K = \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 11 \\ 21 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} = \begin{bmatrix} 11 \times 3 + 8 \times 21 \\ 11 \times 7 + 21 \times 11 \end{bmatrix}$$

$$= \begin{bmatrix} 201 \\ 308 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 19 \\ 22 \end{bmatrix} = \begin{bmatrix} T \\ W \end{bmatrix}$$

$$(iii) \quad P = \begin{bmatrix} E \\ D \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix} \Rightarrow K = \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 4 \\ 3 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} = \begin{bmatrix} 4 \times 3 + 8 \times 3 \\ 4 \times 7 + 11 \times 3 \end{bmatrix}$$

$$= \begin{bmatrix} 36 \\ 61 \end{bmatrix} \pmod{26} = \begin{bmatrix} 10 \\ 9 \end{bmatrix} = \begin{bmatrix} K \\ J \end{bmatrix}$$

Cipher Text: KUTWKJ